

Question		Solution	Marks
2	(a)	The rate of change of momentum of a body is proportional to the resultant force acting on it, and takes place in the direction of the resultant force.	B1 B1
	(b)	(i) $S_x = u_x t + \frac{1}{2} a_x t^2 = (25 \cos 20^\circ) t \quad (a_x = 0)$ $S_y = u_y t + \frac{1}{2} a_y t^2 = (25 \sin 20^\circ) t + \frac{1}{2}(-9.81)t^2$ $\tan 30^\circ = -S_y / S_x$ time = 4.5 s	M1 M1 A1
		(ii) change of velocity $= \sqrt{[(43^2 + 38^2 - 2(43)(38)\cos(24^\circ + 27^\circ))]}$ $= 35 \text{ m s}^{-1}$	M1 A1
		(iii) Resultant force = rate of change of momentum $= m \Delta v / \Delta t$ $= (0.20) (35) / (0.050)$ $= 140 \text{ N}$	A1

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3	(a)			Taking moments about the handle	